

Alexander Lopez

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EDUCATION

Northeastern University, Khoury College of Computer Science

September 2024 – May 2028

- Combined BSE in Computer Science and Mathematics with a Concentration in AI *Cumulative GPA: 3.65/4.0*
- **Programming Languages:** C++ | C | Python | Java | JavaScript | OCaml | Kotlin *Major GPA: 3.84/4.0*
- **Frameworks and Tools:** PyTorch | TensorFlow | React | SQL | AWS | Pandas | HTML/CSS | Three.js | MATLAB
- **Relevant Courses:** Data Structures and Algorithms (Java) | Advanced Probability and Statistics | Number Theory I | Group Theory | Numerical Methods for Differential Equations | Differential Geometry | Discrete Math I

Hopkins School

June 2024

- **Extracurriculars:** Science Olympiad | Editor at Large: The Razor | Varsity Wrestling (Captain), Football, and Track | Chess Club

PROFESSIONAL AND LEADERSHIP EXPERIENCE

Expandya

June 2026 – September 2026

Software Engineering Intern

Boston, MA

- Architected and optimized database systems in Python and C++ to efficiently query and manage datasets exceeding 50M+ rows for eCommerce clients, implementing indexing strategies and query caching that reduced latency by over 1,000%, directly improving site performance and conversion rates under high-concurrency traffic conditions
- Delivered weekly technical presentations to clients, translating complex data insights into actionable eCommerce strategies around site optimization and web performance to drive measurable improvements in customer acquisition and retention

Northeastern University

September 2025 – Present

Teaching Assistant

Boston, MA

- Assist students with coursework in Intro to Databases CS 3200 course covering SQL querying, data modeling, normalization, indexing, transactions, concurrency, that culminates in a final project of building a full-stack database backed applications with APIs and dashboards

O'Hern Lab at Yale University

June 2023 – September 2023

Computational Biology Research Assistant

New Haven, CT

- Engineered a data pipeline to transform raw 3D protein folding simulations from AlphaFold and NetSurfP neural networks into structured, analyzable datasets; optimized Python scripts on Yale's high-performance computing (HPC) clusters to process and classify 10,000+ oncogenic protein variants, identifying which mutations alter folding behavior and structure in ways directly linked to cancer pathogenesis
- Developed MATLAB scripts to visualize and interpret simulation outputs, uncovering structural patterns across pathogenic variant datasets and their downstream effects on protein stability; presented weekly progress updates to the research team and delivered a concluding poster presentation to the Yale Department of Computational Biology on key findings in protein-protein interactions (PPI) and their role in disease-driving mutations

INDEPENDENT CODING PROJECTS

Twitter/X Scraper

June 2026 – Present

- Built a pipeline to collect and analyze public tweets from any X/Twitter profile without the costly official API, persisting scraped data to a server-side database to minimize repeat requests and reduce IP ban exposure, then running the cached data through a three-layer NLP sentiment ensemble combining TextBlob, VADER, and a fine-tuned DistilBERT transformer via HuggingFace and PyTorch to automatically classify each tweet as positive, negative, or neutral

IMC Prosperity 3

March 2025 – April 2025

- Finished top 8% (912 out of ~13,000) in a global algorithmic trading competition; working alongside my teammates, researched and designed various trading opportunities that involved market making, scalping, locational arbitrage, and mean reversion of alpha to optimize both execution and risk-adjusted returns

Coin Shroud

April 2025 – May 2025

- Designed and built a full-stack cryptocurrency investment analytics app as a part of CS 3200, engineering a MySQL data base backed, a middleware API, and a Streamlit frontend to support user roles including investors, developers and regulatory bodies
- Implemented features for live data tracking, portfolio management, and compliance reporting; containerized the entire application using docker and Docker Compose for seamless multi-service deployment

SKILLS AND INTERESTS

Skills: Microsoft Office Suite | Analytical | Communication | Project Management | Problem-Solving | Adaptable | Attention to Detail

Interests: Running (training for my 1st marathon) | Golf | Weightlifting and Nutrition | Chess | Animals (Especially my bunny Luna)